

State Sensitivity Is Not Grounding: Dual Replay for LLM Battery Scheduling

Jiangwei Xue
Independent Researcher
jw.xue@lucentia.company

Zhida Qin
Independent Researcher
shin_robotics@akane.waseda.jp

Yuda Bi*
TReNDS Center (GSU, GaTech, Emory)
yudabi95@gmail.com

DOI: 10.5281/zenodo.22707487

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Abstract

Battery schedules depend on the current state of charge (SOC). A model can react to an SOC change yet produce an irrelevant or unsafe trajectory. We present an interventional audit for direct, one-shot language-model scheduling. Forecasts, prices, constraints, events, and horizon remain fixed while the represented SOC changes by 50, 75, or 100 kWh. Byte-identical repeated requests estimate hosted-call variation. Each raw schedule is replayed from both the represented state and the authoritative plant state. Across 60 challenge scenarios, two hosted Flash conditions showed a mean canonical-stale distance of 466.88 kWh versus 207.13 kWh between duplicate canonical calls. The null-adjusted difference was 259.75 kWh [55.75, 464.80] and was positive in 20/60 scenarios. Under the complete contract, DeepSeek mostly returned zero schedules and Qwen3.6 returned active schedules with widespread SOC and export violations; neither produced an authoritative-feasible schedule in 480 branches. In a separate Qwen3.7-Plus tier, F1 yielded 6/240 carrier-consistent and 3/240 authoritative-feasible schedules, while the F0 payload subset yielded 1/60 and 0/60. Four schedules passed replay under the represented state and failed under the authoritative state. A common deterministic gate repaired every historical F1 branch. Matched zero and deterministic references separated gate-created feasibility from proposal value. An independent checker reproduced all 4620 physical verdicts with zero disagreement. The resulting intervention-to-authority protocol measures state grounding in continuous physical decisions.

Keywords. agent evaluation; battery scheduling; environmental grounding; large language models; physical feasibility; state intervention.

1 Introduction

A grid-connected battery acts from a state that changes with every executed interval. State of charge (SOC), usable capacity, converter limits, reserve commitments, export limits, forecasts, and prices determine which schedules remain admissible. Model predictive control (MPC) makes this dependence explicit by rebuilding the optimization problem from the current state and available information [1–3]. Language models are now being studied as energy interfaces, schedule generators, supervisory agents, and optimization surrogates [4–9]. Recent reviews and power-dispatch frameworks place these systems on a spectrum from direct numerical generation to tool-mediated orchestration [10–12]. This paper asks a narrower question: when a numerical schedule changes after the represented physical state changes, what has actually been measured?

*Corresponding author.

A static feasibility test answers whether one output satisfies one instance. It does not reveal whether the model used the state that made the instance different. Output variation alone also leaves the question open. Hosted calls can vary under the same payload, and a large action change can point away from the physically admissible region. In continuous physical scheduling, state grounding therefore has several observable contracts: the action should respond beyond ordinary repeated-call variation; the response should have a state-consistent direction; the schedule should satisfy the physical contract under the state supplied to the model; and the same schedule should remain valid when replayed from the authoritative plant state.

Several research lines motivate this measurement problem. Direct-optimization benchmarks test whether language models can produce constraint-faithful schedules or optimization solutions under deterministic verification [13–15]. Agent and embodied-AI work studies action grounding, state maintenance, stale evidence, and feedback-driven replanning [16–22]. Repeated-trial benchmarks and stochastic evaluation methods show that reliability cannot be inferred from one run [23–25]. Runtime-assurance research provides separate mechanisms for enforcing physical constraints around learned or experimental controllers [26–28]. The measurement gap lies at their intersection: a continuous physical schedule needs a controlled state intervention, a matched repeat null, and replay under both the represented and authoritative dynamics.

We study that problem in a photovoltaic (PV)–battery environment with 15-minute actions and a 24-hour horizon. The experimental program contains four non-pooled evidence tiers. E2 identifies state response by changing only the physical SOC carrier and comparing canonical–stale distance with a byte-identical canonical duplicate. F1 supplies the complete battery contract and two repeated calls per carrier. F0 repeats a fixed subset of the same payloads after approximately 6.4 days. Qwen3.7-Plus is retained as a later sensitivity tier with separate denominators. Every proposal is scored before clipping, projection, or repair.

The paper makes four contributions.

1. We formalize an intervention-to-authority audit that separates state response, directional consistency, carrier-consistent feasibility, authoritative feasibility, and post-gate proposal value.
2. We combine a controlled 50/75/100-kWh SOC intervention with a byte-identical repeated-call null, identifying state-carrier effects without assuming deterministic hosted inference.
3. We introduce dual replay for physical schedules. The same action is evaluated from the model-facing carrier state and from the authoritative state, exposing authority-transfer failures that a single replay cannot detect.
4. We connect raw-action evidence to a matched execution-gate analysis using zero, heuristic, stale-optimal, random, and same-information MPC references. This locates the source of post-gate feasibility and economic performance.

To our knowledge, prior work on direct scheduling, state grounding, and constraint-faithful agent evaluation has not combined three elements in one physical scheduling protocol: a controlled intervention on a decision-relevant continuous state, an identical-payload repeat null, and deterministic replay under represented and authoritative states [14, 15, 20–22]. The protocol covers open-loop schedule proposals under frozen hosted conditions. Tool-assisted and closed-loop energy-management systems define related but different execution architectures.

2 Related work

2.1 Direct language-model optimization and energy scheduling

Language models have been used in power scheduling as intent translators, direct dispatch agents, residential schedulers, and surrogates for deterministic optimizers. Mongaillard et al. convert user requests into optimization and allocation decisions, while Cheng et al. study an LLM

for advanced power dispatch [4, 6]. Residential and building studies connect language models with home-energy scheduling, MPC, or MILP-derived supervision [5, 7–9]. Other architectures reserve numerical optimization and verification for deterministic modules and use the language model for orchestration or structured parameterization [12, 29]. A recent energy-systems review surveys this broader landscape [10], and a power-dispatch-agent framework emphasizes planning, memory, tool invocation, and action as separate capabilities [11].

Benchmark work increasingly isolates numerical and constraint faithfulness. ConstraintBench asks models to return structured solutions to fully specified optimization problems without solver access and checks every constraint against deterministic references [13]. PowerAtlas evaluates physical feasibility and economic outcomes of power schedules over oracle-labeled instances [14]. SCHEDBench uses matched scheduling instances to test whether semantically equivalent surface forms induce above-noise changes in hard-constraint violations [15]. PlanBench provides a broader planning benchmark grounded in automated-planning domains [30]. Electricity-domain benchmarks extend the scope to factuality, stability, live infrastructure, steady-state studies, and dynamic studies [31–34]. These studies establish feasibility and constraint faithfulness as central evaluation targets. Our protocol focuses on the state dependence of one direct numerical schedule and whether that dependence survives authoritative replay.

2.2 Environmental state and action grounding

Embodied-agent research provides a long line of work on mapping language to admissible actions and updating plans as the environment changes. Zero-shot language-model planners often require explicit grounding to executable actions [16]; SayCan combines language scores with robot affordances [17]; ProgPrompt and Code as Policies constrain or compile language-model outputs into executable robot programs [35, 36]. ReAct and Inner Monologue interleave reasoning with observations or feedback, allowing plans to update after interaction [18, 37]. ALFWorld similarly aligns abstract text environments with embodied tasks [38].

State representation itself has also become an explicit object of study. Statler maintains a structured world state across long-horizon embodied reasoning [19]. GroundAct evaluates whether actions respect environmental prerequisites, while EnvTrustBench and STALE test incorrect, stale, or superseded information [20–22]. CausalWorld shows how controlled interventions on environment variables can support causal and transfer evaluation in robotics [39]. Our setting adds continuous battery dynamics and a counterfactual replay operation: one fixed trajectory can be evaluated from two initial SOC values with identical forecasts, limits, and event information. The intervention is graded in physical units, which supports direction and dose analyses.

2.3 Repeated trials, verification, and execution gates

Agent evaluation has moved beyond single-run success. AgentBench evaluates decision-making across interactive environments, while AgentBoard exposes intermediate progress rather than only final success [40, 41]. τ -bench measures reliability over repeated trials through pass^k [25]; Agent-SafetyBench evaluates safety failures across interactive agent environments [42]. Repeated-execution studies and stochastic evaluation methods likewise recommend distributions or repeat-aware estimators [23, 24, 43]. We use a byte-identical canonical request as an experimental null inside each E2 block, so state-carrier differences are compared with variation observed under the same scheduling payload.

External verification and runtime assurance form a separate lineage. LLM-Modulo combines language models with model-based verifiers [44]. Simplex architectures separate an experimental controller from a trusted safety controller [26]; shielding and predictive safety filters enforce state and input constraints around learned policies [27, 28]. In energy control, MPC and economic MPC offer deterministic references for constrained battery operation [1, 3], while battery scheduling can explicitly trade market value against degradation [45]. The execution

gate in this study belongs to this runtime-assurance tradition. We use it to ask how much of the final feasible trajectory is supplied by the proposal and how much is created by the gate.

3 Physical problem and grounding contracts

3.1 PV–battery dynamics

Each scenario i contains an exogenous suffix

$$z_i = \{\ell_{it}, p_{it}^{\text{pv}}, \lambda_{it}, \theta_{it}\}_{t=0}^{H_i-1}, \quad (1)$$

where ℓ_{it} is load, p_{it}^{pv} is the PV forecast, λ_{it} is energy price, and θ_{it} collects time-varying battery and grid limits. The state includes SOC e_{it} , usable capacity E_{it} , reserve E_{it}^{res} , charge and discharge limits, and the export limit.

The model returns a signed battery schedule $u_i = (u_{i0}, \dots, u_{i,H_i-1})$, with positive values for charging. Let $[a]_+ = \max(a, 0)$. For $\Delta t = 0.25$ h, the deterministic SOC update is

$$e_{i,t+1} = e_{it} + \eta_{\text{ch}}[u_{it}]_+ \Delta t - \frac{[-u_{it}]_+}{\eta_{\text{dis}}} \Delta t. \quad (2)$$

Grid exchange, positive for import, is

$$g_{it} = \ell_{it} - p_{it}^{\text{pv}} + u_{it}. \quad (3)$$

A raw schedule is engineering-feasible when every interval satisfies the signed power bounds, reserve and usable-capacity bounds on $e_{i,t+1}$, and the export condition $g_{it} \geq -P_{it}^{\text{exp}}$, while the terminal state satisfies

$$|e_{i,H_i} - 325| \leq 1 \text{ kWh}. \quad (4)$$

The ± 1 kWh terminal tolerance defines the raw-action endpoint. The deterministic execution gate uses an exact terminal equality. We retain these contracts as separate fields.

Let

$$\Phi_i^\epsilon(u; x, z) \in \{0, 1\} \quad (5)$$

denote deterministic replay from initial state x with terminal tolerance ϵ , and let $\mathbf{v}_i^\epsilon(u; x, z)$ contain the associated power, SOC, export, and terminal violation magnitudes. Binary feasibility and violation severity are both reported.

For post-gate economic comparisons, the physical cost is

$$J_i(u) = \sum_{t=0}^{H_i-1} \left[\lambda_{it} g_{it} \frac{\Delta t}{1000} + c_{\text{deg}} |u_{it}| \Delta t \right]. \quad (6)$$

The prompt supplied prices and the complete physical contract. It did not instruct the model to minimize Eq. (6). Cost and the same-information MPC are evaluation references applied after the proposal is formed.

3.2 Represented and authoritative state

Let x_i^A be the authoritative event-time state reconstructed from the executed prefix. The canonical carrier contains $x_i^C = x_i^A$. The controlled stale carrier contains

$$x_i^S = \mathcal{I}_{\delta_i}(x_i^A), \quad |\delta_i| \in \{50, 75, 100\} \text{ kWh}, \quad (7)$$

with all non-carrier content held fixed. We use x_i^R for the state represented to the model in a given branch. Canonical branches have $x_i^R = x_i^A$; stale branches have $x_i^R = x_i^S$.

For model condition m , carrier $b \in \{C, S\}$, and repeated call r , the proposal is treated as a hosted random output

$$U_{imbr} \sim \Pi_m(\cdot \mid z_i, c(x_i^b)), \quad (8)$$

where $c(\cdot)$ is the frozen state-carrier encoding.

3.3 Five observable contracts

We organize the audit through five observable contracts.

1. Response beyond repeat variation. For two schedules, define the action-energy distance

$$d_i(u, v) = \Delta t \sum_{t=0}^{H_i-1} |u_t - v_t|. \quad (9)$$

E2 contains a byte-identical canonical duplicate $C2$. After reducing three repetitions to a median within each scenario-model cell, the null-adjusted carrier response is

$$\tau_{im} = D_{im}^{CS} - D_{im}^{CC2}. \quad (10)$$

A positive τ_{im} means that the carrier contrast exceeds the observed duplicate-call contrast in that cell. For cross-scenario interpretation we also report

$$\bar{d}_i(u, v) = \frac{d_i(u, v)}{H_i \Delta t}, \quad d_i^\delta(u, v) = \frac{d_i(u, v)}{|\delta_i|}, \quad (11)$$

which express mean absolute power difference and action distance per kilowatt-hour of SOC intervention.

F1 contains two calls per carrier. Its finite-sample two-carrier energy statistic is

$$B_{im} = \frac{1}{4} \sum_{j=1}^2 \sum_{k=1}^2 d_i(C_j, S_k), \quad (12)$$

$$W_{im}^C = d_i(C_1, C_2), \quad W_{im}^S = d_i(S_1, S_2), \quad (13)$$

$$ED_{im} = 2B_{im} - W_{im}^C - W_{im}^S. \quad (14)$$

With two calls per group, ED_{im} can be negative. We report its distribution, scenario support, and robust summaries alongside its mean.

2. Directional consistency. Let $\bar{U}_i^C$ and $\bar{U}_i^S$ be the two-call carrier means in F1, and let $\delta_i = e_i^C - e_i^S$. A coarse directional score is

$$R_i^{\text{dir}} = -\text{sgn}(\delta_i) \Delta t \sum_t (\bar{U}_{it}^C - \bar{U}_{it}^S). \quad (15)$$

Positive values indicate that a lower represented SOC leads to more net charging or less net discharging. This score measures the sign of aggregate energy response; it does not encode the full time-varying feasible set.

3. Carrier-consistent feasibility. The proposal is replayed from the model-facing state:

$$F_{imbr}^{\text{carrier}} = \Phi_i^1(U_{imbr}; x_i^R, z_i). \quad (16)$$

This endpoint asks whether the schedule is valid in the state represented to the model.

4. Authoritative feasibility. The same proposal is replayed from the plant state:

$$F_{imbr}^{\text{auth}} = \Phi_i^1(U_{imbr}; x_i^A, z_i). \quad (17)$$

An authority-transfer failure is

$$L_{imbr}^{\text{auth}} = \mathbf{1}\{F_{imbr}^{\text{carrier}} = 1, F_{imbr}^{\text{auth}} = 0\}. \quad (18)$$

We use F^{auth} as the safety-facing raw endpoint.

5. Proposal value after a common gate. Let $G_i(u)$ solve a deterministic feasibility-restoration problem from authoritative state. The gate may alter battery power and, in the expanded formulation, use curtailment and mode variables. Its post-gate regret is

$$\text{Regret}_i(G_i(u)) = J_i(G_i(u)) - J_i(u_i^*), \quad (19)$$

where u_i^* is the same-information MPC reference. Matched proposal value against baseline b is measured by

$$\Delta \text{Regret}_{m,b} = \text{Regret}_i(G_i(U_m)) - \text{Regret}_i(G_i(U_b)). \quad (20)$$

A feasible $G_i(U_m)$ establishes feasibility of the proposal-gate system. Eq. (20) locates any incremental economic value supplied by the proposal.

4 Experimental design

4.1 Public-data substrate and plant

We assemble the deterministic environment from public 2024 time series at 15-minute resolution. PV input uses NSRDB GOES CONUS v4 data for Los Angeles, San Diego, Fresno, and Sacramento [46]; `pvl` converts irradiance and weather to AC power [47]. Load follows the OEDI ComStock LargeOffice profile [48, 49]. Price is CAISO day-ahead locational marginal price at TH_NP15_GEN-APND [50]. The synchronized series define a controlled benchmark substrate; they are not measurements from a co-located plant.

The simulated site combines a 400-kW PV array, a commercial load scaled to an 800-kW peak, and a 500-kWh battery. The battery accepts up to 250 kW in either direction. Both one-way efficiencies are 0.95; the default reserve and export limits are 75 kWh and 150 kW. The battery-throughput coefficient in Eq. (6) is 0.008 USD/kWh; degradation-aware battery optimization motivates including throughput cost [45]. A day contains 96 intervals. Each scenario contains one visible event from five families: forecast revision, battery-capacity derating, power-limit derating, export-limit update, and reserve-commitment update.

4.2 Challenge-set construction

The candidate generator produced 280 E2 scenarios. Provider-free admission required a feasible canonical deterministic suffix, an infeasible zero suffix, at least 25 kWh difference between canonical and stale deterministic plans, and a material consequence such as stale-plan infeasibility, terminal error above 10 kWh, or a first-two-hour action change of at least 20 kW. Two hundred five candidates passed admission, and 60 were selected with 12 scenarios per event family. The set is designed to expose state-dependent failures. Its denominators describe this challenge set. The authoritative SOC is lower than the stale carrier in 37 scenarios and higher in 23. Signed gaps are -100 , -75 , -50 , $+50$, and $+100$ kWh.

4.3 Evidence tiers

Table 1 summarizes the frozen tiers. They share the physical substrate and replay code but answer different questions. Their results are reported separately.

E2 carrier identification. Each of the 60 scenarios is crossed with DeepSeek-V4-Flash and Qwen3.6-Flash-2026-04-16 and run three times, producing 360 blocks. Each block begins with one shared prefix and then forks into five Stage-2 branches: canonical state (C), controlled stale physical state (S), metadata control (K), model-maintained carrier (M), and a byte-identical canonical duplicate (C2). The planned run contained 2160 calls; 2158 were attempted and 2146 returned successfully, with no protocol exclusions. C–S is the physical intervention contrast and C–C2 is the repeated-call null used in the main estimator.

Table 1: Experimental tiers and inferential roles. Calls are logical hosted requests.

Tier	Scenarios	Model conditions	Calls	Primary role
E2	60	2	2,160 planned	Carrier response against a byte-identical null; three repetitions
F1	60	2	480	Complete physical contract; two canonical and two stale calls
F0	15	2	120	Byte-identical temporal rerun of an F1 subset after 6.44 days
Qwen3.7 F1	60	1	240	Separate-model sensitivity tier under the F1 contract
Qwen3.7 F0 payloads	15	1	60	Separate sensitivity on the frozen F0 payload subset

F1 complete-contract tier. F1 uses the same 60 scenario identities. Each scenario–model block contains C1, C2, S1, and S2, for 480 calls. The prompt gives the signed power convention, SOC dynamics, efficiencies, interval-specific charge/discharge, reserve, capacity and export limits, the full load/PV/price suffix, and the 325-kWh terminal target with a 1-kWh engineering tolerance. Output is a sparse JSON segment list that expands deterministically to the dense action vector. Temperature is zero, provider reasoning is disabled where available, and no tools, solver, search, code execution, or fallback route is permitted.

F0 temporal rerun. F0 resubmits 120 byte-identical payloads from 15 scenarios per historical model condition. The first F0 request starts 154.65 hours after the last F1 request. This tier measures paired temporal reproducibility across the two saved service windows.

Qwen3.7-Plus sensitivity. A later tier applies the same frozen F1 and F0 payload sets to Qwen3.7-Plus. It contains 240 F1 branches and 60 F0-subset branches. These rows are never pooled with the two historical Flash conditions.

4.4 Reduction and uncertainty

Within each scenario–model cell, E2 takes the median of three repetition-level observations. Model estimates average the resulting 60 cell medians. For the pooled estimate, the two model-cell medians are averaged within scenario before averaging over the 60 scenarios. Parser failure receives zero distance under the registered full-denominator rule; joint parseability is reported beside the estimate. Uncertainty uses 10,000 scenario-cluster resamples with seed 20260716. The resulting intervals measure stability to the composition of the frozen challenge set.

F1 treats the scenario–model block as the independent unit. We report means, medians, positive/ zero/negative support, 10% and 20% trimmed means, and deletion of the largest one, two, and five scenario effects. A random carrier-label pairing test provides an additional finite-sample null for Eq. (14). Dose and direction are stratified by signed SOC gap and model.

4.5 Replay, gate, and evidence validation

Every raw response, parsed action, state carrier, replay, and gate trajectory is hash linked. Raw replay is performed before any repair. The F1 battery-only gate seeks the minimum same-action-space correction under exact terminal equality; the expanded gate can also use curtailment and mode variables. Global distance optimality is certified for 409/480 historical F1 branches. The remaining 71 branches retain a feasible mode-library upper bound on correction distance. The same gate is applied to language-model, zero, heuristic, random static-range, stale-optimal, and direct-MPC proposals.

A second physics checker independently implements SOC transition, power limits, reserve, usable capacity, export, curtailment, terminal semantics, and cost. It imports no simulator, scorer, or gate functions from the primary implementation. The release compares 4620 saved raw, gated, and deterministic-witness trajectories and includes 12 hand-specified boundary fixtures.

5 Results

5.1 The SOC carrier changes some schedules beyond the repeat null

Table 2 reports the E2 identification result. The mean canonical-stale distance was 466.88 kWh [280.24, 671.80]. Byte-identical canonical calls differed by 207.13 kWh [88.75, 344.00]. Their null-adjusted contrast was 259.75 kWh [55.75, 464.80]. Joint parsing succeeded for C/S in 234/360 blocks and for C/C2 in 238/360.

Table 2: E2 action-distance results. Values are kWh; brackets are scenario-cluster resampling stability intervals.

Estimate	Mean	Stability interval
Canonical-stale, pooled	466.88	[280.24, 671.80]
Canonical-duplicate, pooled	207.13	[88.75, 344.00]
Null-adjusted, pooled	259.75	[55.75, 464.80]
Null-adjusted, DeepSeek-V4-Flash	405.58	[120.99, 704.76]
Null-adjusted, Qwen3.6-Flash	113.91	[-178.91, 426.41]

The model strata differ. DeepSeek’s adjusted interval stays above zero; Qwen3.6’s interval spans zero. At scenario level, the pooled adjusted value is positive in 20/60 cases, zero in 34/60, and negative in 6/60. Capacity and power derating carry the largest event-family means, while forecast and export updates are centered near zero. The aggregate effect reflects a subset of state-active cases rather than a uniform response across the challenge set.

5.2 Complete-contract responses are sparse and model-specific

F1 changes the interface from E2’s multi-branch experiment to a complete, direct schedule contract with two calls per carrier. Table 3 shows three distinct output phenotypes. DeepSeek returned 218/240 exact-zero schedules; Qwen3.6 returned active schedules in all 240 branches; Qwen3.7-Plus was active in 239/240 branches. Their response distributions differ as much as their action levels.

Table 3: F1 response and authoritative-replay outcomes by model condition. Qwen3.7-Plus is a separate sensitivity tier. ED support is positive/zero/negative over 60 scenario blocks.

Model condition	Nontrivial (/240)	Mean ED (kWh)	Median ED (kWh)	ED support (+/0/-)	Dynamic OK (/240)	Terminal viol. (/240)	F^{carrier} (/240)	F^{auth} (/240)
DeepSeek-V4-Flash	22	1173.33	0	8/52/0	223	240	0	0
Qwen3.6-Flash	240	345.45	0	3/56/1	1	240	0	0
Qwen3.7-Plus	239	1461.55	154.40	37/16/7	81	237	6	3

DeepSeek’s mean ED was 1173.33 kWh, yet its median was zero and only 8/60 scenarios were positive. Removing the five largest scenarios reduced the mean to 91.36 kWh; the 20% trimmed mean was zero. Qwen3.6 had a mean ED of 345.45 kWh, a zero median, and only three positive scenarios. Removing its largest scenario moved the mean below zero. Random carrier-label pairing placed the DeepSeek mean above its null distribution (upper-tail $p = 0.00018$); the Qwen3.6 mean did not ($p = 0.13536$). Qwen3.7-Plus produced broader support, with a 154.40-kWh median and 37 positive scenarios, while still retaining seven negative scenario statistics.

After division by horizon duration, mean ED is 51.56 kW for DeepSeek, 16.65 kW for Qwen3.6, and 64.86 kW for Qwen3.7-Plus; the corresponding medians are 0, 0, and 6.80 kW.

The two historical Flash conditions fail in different ways. DeepSeek usually submits no action; its dynamic constraints pass in 223/240 branches, but every branch misses the terminal target. Qwen3.6 always acts, and 239/240 branches violate at least one dynamic constraint. SOC lower-bound violations occur in 235 branches and export violations in 138. Both models finish with 0/240 authoritative-feasible schedules.

5.3 Larger SOC interventions do not produce a monotone response

Table 4 stratifies historical F1 by the absolute SOC gap. Every model-dose median is zero. DeepSeek’s mean ED rises from 50 to 75 kWh and then falls at 100 kWh. Qwen3.6 changes sign at 50 kWh and has its largest mean at 100 kWh. Directional support also remains sparse. The data do not show a monotone dose-response relation over the three registered gaps. Event-family composition differs across gap strata, so this table describes the registered challenge set rather than an isolated causal dose effect.

Table 4: Historical F1 response by SOC intervention magnitude. Direction support counts positive, zero, and negative coarse directional scores over 20 scenarios per cell.

Model	Gap (kWh)	Mean ED (kWh)	Median ED	ED support	Direction support
DeepSeek-V4-Flash	50	1137.50	0	2/18/0	1/18/1
	75	1975.00	0	3/17/0	2/17/1
	100	407.50	0	3/17/0	4/16/0
Qwen3.6-Flash	50	-155.25	0	0/19/1	2/15/3
	75	126.50	0	1/19/0	1/13/6
	100	1065.10	0	2/18/0	2/16/2

Across all gaps, DeepSeek’s coarse directional score was positive in 7/60 scenarios, zero in 51, and negative in two. Qwen3.6 was positive in five, zero in 44, and negative in 11. Positive action distance alone provides little evidence that the response follows the sign of the SOC intervention.

5.4 Carrier-consistent feasibility can fail under authoritative replay

Table 5 applies the two replay states to every complete-contract branch. The historical Flash tiers remain infeasible under both definitions. Qwen3.7-Plus changes the picture: 6/240 F1 schedules are feasible from the state represented to the model, while 3/240 remain feasible from the authoritative state. On the 60 F0-subset payloads, one schedule is carrier-consistent and none is authoritative-feasible.

Table 5: Raw feasibility under the represented carrier state and authoritative state. Tiers are kept separate.

Tier	Branches	Carrier-consistent	Authoritative
Historical F1, two Flash conditions	480	0	0
Historical F0 temporal rerun	120	0	0
Qwen3.7-Plus F1	240	6	3
Qwen3.7-Plus F0 payload subset	60	1	0

The four authority-transfer failures are all stale-carrier schedules. Three appear in Qwen3.7 F1 and one in its F0-subset tier. Their carrier-state terminal errors range from 0.03 to 0.71 kWh, inside the engineering tolerance. Authoritative replay changes those errors to approximately 49.97, 50.19, 75.71, and 99.81 kWh. No other constraint creates these four mismatches. The fixed action is internally valid for the represented initial SOC and invalid for the plant initial SOC.

5.5 The temporal rerun preserves the feasibility conclusion

All 120 historical F0 request payload hashes match their F1 counterparts. DeepSeek repeats 59/60 branch actions exactly and 14/15 four-branch blocks exactly. Qwen3.6 repeats 49/60 branch actions and 9/15 blocks. Raw authoritative feasibility remains 0/120. This paired rerun shows that output reproducibility is model-dependent even at temperature zero, while the physical endpoint is unchanged across the two service windows.

5.6 A common gate separates feasibility from proposal value

The exact-terminal battery-only gate solves all 480 historical F1 branches. Post-gate feasibility is 480/480, while raw authoritative feasibility is 0/480. The required corrections are model-specific. DeepSeek’s median correction is 131.58 kWh and Qwen3.6’s is 5221.84 kWh. The median post-gate regret is 3.12 USD for each model. The common median reflects the gate and scenario set more than the raw action distribution.

Table 6 reports matched differences against a zero proposal under the same gate. DeepSeek has zero median differences in correction and regret; 218/240 correction differences and 224/240 regret differences are exactly zero. Qwen3.6 requires 5187.50 kWh more median correction than zero. Its mean regret is 0.126 USD lower than zero, but the median is zero and the difference is confined to 8/240 branches. The gate supplies feasibility in every branch. The raw proposal supplies limited and comparator-dependent economic information.

Table 6: Matched LLM-minus-zero differences under the battery-only gate. Support is positive/zero/negative. Negative regret favors the LLM proposal. Intervals are scenario-resampling stability intervals.

Model	Mean ΔD (kWh)	Median ΔD	ΔD support	Mean Δregret (USD)	Median	Regret support
DeepSeek-V4-Flash	337.58 [187.74, 510.46]	0	17/218/5	0.253 [-0.330, 1.006]	0	9/224/7
Qwen3.6-Flash	4633.93 [4489.48, 4774.67]	5187.50	240/0/0	-0.126 [-0.245, -0.035]	0	0/232/8

The expanded gate uses curtailment in 8/240 branches for each historical model. Most rescues alter battery action and charge/discharge mode: 227/240 DeepSeek branches and 231/240 Qwen3.6 branches. These counts show that repair can include control variables absent from the language-model action space. Post-gate feasibility belongs to the full execution layer.

5.7 Independent replay closes the physical evidence loop

The independent checker agrees with the primary implementation on all 4620 saved comparisons: 1,800 raw replay rows, 2,760 gate trajectories, and 60 deterministic witnesses. All 12 hand-specified fixtures pass, including power, SOC upper and lower bounds, export, terminal tolerance, reserve, capacity derating, and boundary cases. The maximum numerical difference is zero at the stored precision, and no Boolean feasibility verdict differs.

6 Discussion

6.1 State grounding is a sequence of contracts

The experiments reveal distinct points of failure. E2 detects a carrier effect beyond an observed repeat null, yet that effect is concentrated in a minority of scenarios and differs by model. F1 shows that response magnitude can have a zero median and a large mean. Directional support is sparser still. Qwen3.7-Plus provides the final separation: several schedules satisfy the complete contract from the represented state, while four lose feasibility under authoritative replay.

These results support a contract view of state grounding. A physically grounded schedule should show state-dependent behavior beyond repeat variation, align with the direction of the

intervention, and retain validity under the state that the plant will execute. Each contract has a different evidence object. Response uses paired raw actions; direction uses signed intervention; feasibility uses deterministic replay; authority transfer requires the same action under two initial states. A single success rate would hide these distinctions.

6.2 The repeat null changes the interpretation of sensitivity

The E2 byte-identical distance is 44% of the pooled canonical–stale distance. Treating identical calls as deterministic would assign that entire difference to SOC. The within-block duplicate provides a direct service-window reference. It also exposes model heterogeneity: DeepSeek’s adjusted interval stays above zero, while Qwen3.6’s does not.

F1 reinforces the need for distributional reporting. DeepSeek’s 1173.33-kWh mean ED is supported by eight scenarios and a zero median. Qwen3.6 has one negative and three positive scenarios. These patterns resemble thresholded or episodic state use more than a smooth controller response. The 50/75/100-kWh strata do not yield a monotone curve.

6.3 Dual replay identifies a deployment-relevant failure

Carrier-consistent replay measures internal validity under the state presented to the model. Authoritative replay measures the action that would reach the plant. The four Qwen3.7 authority losses are especially clear because the schedule has no nonterminal violation in either replay; the initial SOC shift alone moves the terminal state outside tolerance. A conventional single replay would classify these rows according to whichever initial state the evaluator happened to choose.

The distinction applies beyond batteries. Any physical agent that receives an estimated, cached, or model-maintained state can produce an action that is valid in that representation and invalid in the execution state. Dual replay offers a general evaluation pattern when the dynamics are deterministic enough to reconstruct both trajectories.

6.4 Execution gates need matched attribution

The F1 gate restores feasibility for every historical branch, including zero proposals and highly active infeasible proposals. Its output combines the raw action, the exact-terminal contract, the lexicographic objective, and additional variables such as curtailment and mode. The matched zero comparison shows how little of the final result can be inferred from post-gate feasibility alone. DeepSeek largely reproduces the zero baseline. Qwen3.6 occasionally changes cost by a small amount, while requiring several megawatt-hours of additional trajectory correction.

This suggests a practical division of labor. Authoritative state belongs in a plant-side estimator or digital twin. Deterministic optimization owns modeled feasibility. Language models can serve as interfaces, intent translators, or proposal generators, as in solver-mediated and orchestration-focused energy architectures [12, 29]. Their contribution should be measured against simple proposals under the same execution layer. This comparison is more informative than reporting a repaired success rate.

6.5 Implications for benchmark design

A physical-agent benchmark gains diagnostic power from four design choices. First, intervene on a state variable whose physical consequence is known. Second, include identical-payload repeats in the same service window. Third, preserve the represented state and the authoritative state as separate replay inputs. Fourth, store raw and gated actions with stable row-level identifiers. These choices support causal interpretation at the protocol level without requiring a model-based judge.

The protocol also favors endpoint families over composite scores. Parseability, response, direction, violation severity, authoritative feasibility, correction distance, and regret answer

different questions. Reporting them together preserves the path from model output to physical execution.

7 Limitations

The 60 scenarios form a purpose-built challenge set. Admission selects state-sensitive, constraint-active cases, so the observed feasibility rates do not estimate ordinary PV–battery operation. The candidate archive preserves selection metadata. Complete PV, load, and price vectors were not retained for every rejected candidate, so a full counterfactual reselection cannot be reconstructed.

The main hosted conditions are DeepSeek-V4-Flash and Qwen3.6-Flash under direct, one-shot, structured-output prompts. Qwen3.7-Plus is a separate sensitivity tier. Reasoning tools, code execution, solver calls, iterative correction, constrained decoding beyond JSON output, and closed-loop replanning are outside the experiment. The results characterize schedule-proposal reliability under these contracts.

E2 has three repetitions per scenario–model cell and a byte-identical null. F1 has two calls per carrier, which makes Eq. (14) variable and permits negative estimates. We report support, trimmed summaries, deletion analyses, and a pairing null, but more repetitions would estimate the full hosted-output distribution more precisely.

The prompt specifies feasibility and the terminal target. It does not specify Eq. (6) as the optimization objective. We exclude MPC trajectory distance as a measure of model optimality. Post-gate regret measures the execution system produced from each proposal.

The substrate synchronizes public PV, load, and price data from different sources and is not a co-located field installation. The model proposes one open-loop suffix after one event. Sensor error, battery ageing, network power flow, demand charges, hardware timing, receding-horizon execution, and field protection remain untested.

The archived E2 projection population retains 53/1800 timeouts and has status `PARTIAL_WITH_TIMEOUTS`. It is excluded from the complete-denominator gate claims in this paper. The matched F1 gate population used in Section 5 solves all 480 branches.

8 Conclusion

This study evaluates whether direct language-model battery schedules use physical state in a form that survives execution. A controlled SOC intervention changes some proposals beyond an identical-payload repeat null, but the response is sparse, model-dependent, and non-monotone over 50, 75, and 100 kWh. The historical Flash conditions produce no authoritative-feasible schedule under the complete contract. Qwen3.7-Plus produces a small set of carrier-consistent feasible schedules, four of which fail when replay begins from the authoritative state. A deterministic gate restores feasibility, while matched baselines show that the gate and the proposal contribute different parts of the final result.

The central evaluation object is the path from state intervention to authoritative replay. This path separates action sensitivity, directional use of state, physical validity in the represented world, physical validity at the plant, and downstream proposal value. The resulting protocol is applicable to other continuous physical agents whenever actions can be replayed under both represented and authoritative dynamics.

Data and code availability

Data and code supporting the findings are available in the accompanying reproducibility repository at <https://github.com/Jiangwei-Xue/paper-05-llm-battery-state-grounding>.

The repository provides the frozen tasks, prompts and request payloads, raw hosted-model responses, parsed actions, deterministic dual replay, scores, branch comparisons, matched gate baselines, claim-level hashes, analysis code, reference outputs, and a one-command offline workflow that regenerates the reported analyses from saved provider records and validates the VCR/H5/E5 evidence chain. The workflow does not repeat nondeterministic hosted-model calls. The analysis code is released under the MIT License. Original project documentation and derived result tables are licensed under CC BY 4.0. Third-party data, hosted-model records, and bundled dependencies remain subject to their respective terms.

Competing interests

The authors declare no competing interests.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

A Frozen schedule prompt contract

The complete-contract prompt asks for one JSON object with a single `actions` field. Each segment contains `start_offset`, `end_offset_exclusive`, and `power_kw`. Segments must be sorted and non-overlapping; uncovered offsets expand to zero. Positive power charges the battery and negative power discharges it. The prompt states Eq. (2), all interval-specific power, reserve, capacity and export limits, Eq. (3), and the terminal condition in Eq. (4). It also states that raw actions are scored before clipping, projection, or repair. The task JSON contains the carrier state and complete suffix time series. Explanations and fields outside the schema are disallowed.

B Event-family heterogeneity in E2

Table 7 reports post-hoc event-family estimates. Each row contains 12 scenario clusters. The intervals describe challenge-set resampling stability.

Table 7: E2 null-adjusted action distance by event family.

Event family	Mean (kWh)	Stability interval	Support (+/0/-)
Battery-capacity derating	831.7	[221.6, 1425.8]	8/3/1
Export-limit update	-67.5	[-312.5, 156.7]	2/8/2
Forecast revision	-1.0	[-550.0, 557.3]	2/8/2
Power-limit derating	421.0	[133.1, 734.4]	5/6/1
Reserve-commitment update	114.6	[0.0, 264.6]	3/9/0

C Authority-transfer witnesses

Table 8 lists the four schedules with $F^{\text{carrier}} = 1$ and $F^{\text{auth}} = 0$. All are Qwen3.7-Plus stale-carrier outputs. The signed gap is canonical SOC minus stale SOC.

Table 8: Saved authority-transfer failures. Terminal errors are absolute kWh.

Tier	Branch	Event	Location/season	Signed SOC gap	Carrier error	Authoritative error
F1	S1	Forecast revision	Sacramento/winter	-75	0.71	75.71
F1	S1	Export-limit update	Fresno/spring	+100	0.19	99.81
F1	S1	Capacity derating	Sacramento/autumn	-50	0.19	50.19
F0 subset	S2	Export-limit update	San Diego/autumn	-50	0.03	49.97

D Deterministic reference baselines

Table 9 gives aggregate battery-only gate outcomes for the five deterministic proposal sources over the 60 F1 scenarios. These rows use the same authoritative state and exact terminal gate. Direct MPC is the zero-regret reference by construction.

Table 9: Deterministic proposal baselines under the common battery-only gate.

Proposal source	Raw feasible (/60)	Median correction (kWh)	Median cost (USD)	Median regret (USD)
Direct MPC	60	0.00	337.38	0.00
Simple heuristic	60	0.00	354.47	5.15
Stale-optimal plan	0	71.25	337.38	0.00
Zero schedule	0	131.58	353.16	3.31
Random static-range schedule	0	368.50	361.49	27.23

References

- [1] Alessandra Parisio, Evangelos Rikos, and Luigi Glielmo. A model predictive control approach to microgrid operation optimization. *IEEE Transactions on Control Systems Technology*, 22(5):1813–1827, 2014. doi: 10.1109/TCST.2013.2295737.
- [2] A. L. Cedeño, R. López Ahuar, J. Rojas, G. Carvajal, C. Silva, and J. C. Agüero. Model predictive control for photovoltaic plants with non-ideal energy storage using mixed integer linear programming. *Energies*, 15(17):6427, 2022. doi: 10.3390/en15176427.
- [3] Brendan Banfield, Duane A. Robinson, and Ashish P. Agalgaonkar. Comparison of economic model predictive control and rule-based control for residential energy storage systems. *IET Smart Grid*, 2020. doi: 10.1049/iet-stg.2020.0090.
- [4] Thomas Mongaillard, Samson Lasaulce, Othman Hicheur, Chao Zhang, Lina Bariah, Vineeth S. Varma, Hang Zou, Qiyang Zhao, and Merouane Debbah. Large language models for power scheduling: A user-centric approach, 2024.
- [5] Fabien Michelon, Yuzhou Zhou, and Thomas Morstyn. Large language model interface for home energy management systems. In *Proceedings of the 16th ACM International Conference on Future and Sustainable Energy Systems*, pages 590–602, 2025. doi: 10.1145/3679240.3734586.
- [6] Yuheng Cheng, Huan Zhao, Xiyuan Zhou, Junhua Zhao, Yuji Cao, Chao Yang, et al. A large language model for advanced power dispatch. *Scientific Reports*, 15:8925, 2025. doi: 10.1038/s41598-025-91940-x.
- [7] R. El Makroum, S. Zwickl-Bernhard, and L. Kranzl. Agentic ai home energy management system: A large language model framework for residential load scheduling. *Results in Engineering*, 29:109857, 2026. doi: 10.1016/j.rineng.2026.109857.
- [8] V. S. Raghavan and A. V. Giridhar. Cost-optimal residential energy scheduling via a zero-shot llm agent and model predictive control. *Journal of Building Performance Simulation*, 19(3):492–510, 2026. doi: 10.1080/19401493.2025.2605465.
- [9] Rouzbeh Haghighi, Van-Hai Bui, and Wencong Su. Fine-tuned large language models as surrogate schedulers for smart building energy management. *Energy and Buildings*, 364:117588, 2026. doi: 10.1016/j.enbuild.2026.117588.

- [10] Chaobo Zhang, Jian Zhang, Jie Lu, Yang Zhao, et al. Large language models meet energy systems: Opportunities, challenges, and future perspectives. *Applied Energy*, 403:127076, 2026. doi: 10.1016/j.apenergy.2025.127076.
- [11] Huan Zhao, Yuheng Cheng, Dejun Xiang, Xiyuan Zhou, Junhua Zhao, Xinlei Cai, and Zhaoyang Dong. Large language model-based power dispatch agent: Framework, application and challenges. *International Journal of Electrical Power & Energy Systems*, 175:111653, 2026. doi: 10.1016/j.ijepes.2026.111653.
- [12] Junpeng Zhao, Rouyi Chen, Hui Jiang, Yanlu Huang, et al. An llm-agent-based framework for calculating nodal carbon intensity in regional power systems. *Scientific Reports*, 2026. doi: 10.1038/s41598-026-60053-4.
- [13] J. Tso, P. Schmittou, Q. Huynh, and J. Hutchins. Constraintbench: Benchmarking llm constraint reasoning on direct optimization, 2026.
- [14] K. Jiang, S. Xu, Z. Zhu, C. Yang, A. T. Luu, and H. Luo. Poweratlas: Towards electricity-computing co-scheduling for power systems, 2026.
- [15] Shrenil Shaun Sharma and Avi Sharma. Schedbench: A benchmark for evaluating llm constraint faithfulness in natural-language combinatorial scheduling, 2026.
- [16] Wenlong Huang, Pieter Abbeel, Deepak Pathak, and Igor Mordatch. Language models as zero-shot planners: Extracting actionable knowledge for embodied agents. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 9118–9147, 2022. URL <https://proceedings.mlr.press/v162/huang22a.html>.
- [17] Michael Ahn, Anthony Brohan, Noah Brown, et al. Do as i can, not as i say: Grounding language in robotic affordances, 2022.
- [18] Wenlong Huang, Fei Xia, Ted Xiao, Harris Chan, Jacky Liang, Pete Florence, Andy Zeng, Jonathan Tompson, Igor Mordatch, Yevgen Chebotar, Pierre Sermanet, Tomas Jackson, Noah Brown, Linda Luu, Sergey Levine, Karol Hausman, and Brian Ichter. Inner monologue: Embodied reasoning through planning with language models. In *Proceedings of the 6th Conference on Robot Learning*, volume 205 of *Proceedings of Machine Learning Research*, pages 1769–1782, 2023. URL <https://proceedings.mlr.press/v205/huang23c.html>.
- [19] Takuma Yoneda, Jiading Fang, Peng Li, Huanyu Zhang, Tianchong Jiang, Shengjie Lin, Ben Picker, David Yunis, Hongyuan Mei, and Matthew R. Walter. Statler: State-maintaining language models for embodied reasoning. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024. doi: 10.1109/ICRA57147.2024.10610634.
- [20] Z. Wang, D. Li, H. Li, Y. Miao, S. Chen, Y. Yan, W. Zhang, Y. Shen, W. Lu, J. Xiao, and Y. Zhuang. Groundact: Can llm agents ground actions in environmental states?, 2025. Revised 2026.
- [21] S. Sheng, Z. Wang, and L. Zhou. When agents overtrust environmental evidence: An extensible agentic framework for benchmarking evidence-grounding defects in llm agents, 2026.
- [22] H. Chao, Y. Bai, R. Sheng, T. Li, and Y. Sun. Stale: Can llm agents know when their memories are no longer valid?, 2026.
- [23] J. Cui and R. Alexander. Same prompt, different outcomes: Evaluating the reproducibility of data analysis by llms, 2026.
- [24] G. Lior, E. Habba, S. Levy, A. Caciularu, and G. Stanovsky. Reliableeval: A recipe for stochastic llm evaluation via method of moments. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 11146–11153, 2025. doi: 10.18653/v1/2025.findings-emnlp.594.
- [25] Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. τ -bench: A benchmark for tool-agent-user interaction in real-world domains, 2024.

- [26] D. Seto, B. Krogh, L. Sha, and A. Chutinan. The simplex architecture for safe on-line control system upgrades. In *Proceedings of the 1998 American Control Conference*, volume 6, pages 3504–3508, 1998. doi: 10.1109/ACC.1998.703255.
- [27] Mohammed Alshiekh, Roderick Bloem, Ruediger Ehlers, Bettina Könighofer, Scott Niekum, and Ufuk Topcu. Safe reinforcement learning via shielding. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, pages 2669–2676, 2018. doi: 10.1609/aaai.v32i1.11797.
- [28] Kim P. Wabersich and Melanie N. Zeilinger. A predictive safety filter for learning-based control of constrained nonlinear dynamical systems. *Automatica*, 129:109597, 2021. doi: 10.1016/j.automatica.2021.109597.
- [29] Shihong Su, Jingjie Huang, Manting Wu, and Zhaoyang Dong. From unstructured information to structured optimization: A large language model based framework for microgrid contingency energy management. *Sustainable Energy, Grids and Networks*, 47:102343, 2026. doi: 10.1016/j.segan.2026.102343.
- [30] Karthik Valmeekam, Matthew Marquez, Alberto Olmo, Sarath Sreedharan, and Subbarao Kambhampati. Planbench: An extensible benchmark for evaluating large language models on planning and reasoning about change, 2023.
- [31] X. Zhou, H. Zhao, Y. Cheng, Y. Cao, G. Liang, G. Liu, W. Liu, Y. Xu, and J. Zhao. Elecbench: A power dispatch evaluation benchmark for large language models, 2024.
- [32] E. Curcio. Energyagentbench: Benchmarking llm agents on live energy infrastructure data, 2026.
- [33] C. Mylonas, M. Foti, A. Pomarico, M. Duarte, Q. Zhang, and E. Varvarigos. Poweragentbench-ss: A benchmark for agentic ai in power system steady-state studies, 2026.
- [34] Q. Zhang, A. Pomarico, C. Mylonas, M. Foti, A. Berizzi, and L. Xie. Poweragentbench-dyn: A benchmark for agentic ai in power system dynamic studies, 2026.
- [35] Ishika Singh, Valts Blukis, Arsalan Mousavian, Ankit Goyal, Danfei Xu, Jonathan Tremblay, Dieter Fox, Jesse Thomason, and Animesh Garg. Progprompt: Generating situated robot task plans using large language models. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pages 11523–11530, 2023. doi: 10.1109/ICRA48891.2023.10161317.
- [36] Jacky Liang, Wenlong Huang, Fei Xia, Peng Xu, Karol Hausman, Brian Ichter, Pete Florence, and Andy Zeng. Code as policies: Language model programs for embodied control, 2023.
- [37] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*, 2023. URL https://openreview.net/forum?id=WE_vluYUL-X.
- [38] Mohit Shridhar, Xingdi Yuan, Marc-Alexandre Côté, Yonatan Bisk, Adam Trischler, and Matthew Hausknecht. Alfworld: Aligning text and embodied environments for interactive learning. In *International Conference on Learning Representations*, 2021. URL <https://openreview.net/forum?id=0IOX0YcCdTn>.
- [39] Ossama Ahmed, Frederik Träuble, Anirudh Goyal, Alexander Neitz, Yoshua Bengio, Bernhard Schölkopf, Manuel Wüthrich, and Stefan Bauer. Causalworld: A robotic manipulation benchmark for causal structure and transfer learning. In *International Conference on Learning Representations*, 2021. URL https://openreview.net/forum?id=SK7Ph_iSD2.
- [40] Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, et al. Agentbench: Evaluating llms as agents. In *International Conference on Learning Representations*, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/e9df36b21ff4ee211a8b71ee8b7e9f57-Abstract-Conference.html.
- [41] Chang Ma, Junlei Zhang, Zhihao Zhu, Cheng Yang, Yujiu Yang, Yaohui Jin, Zhenzhong Lan, Lingpeng Kong, and Junxian He. Agentboard: An analytical evaluation board of multi-turn llm agents. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-2365.

- [42] Zhexin Zhang, Shiyao Cui, Yida Lu, Jingzhuo Zhou, Junxiao Yang, Hongning Wang, and Minlie Huang. Agent-safetybench: Evaluating the safety of llm agents, 2024.
- [43] Percy Liang, Rishi Bommasani, Tony Lee, et al. Holistic evaluation of language models. *Transactions on Machine Learning Research*, 2023. URL <https://openreview.net/forum?id=i04LZibEqW>.
- [44] Subbarao Kambhampati, Karthik Valmeekam, Lin Guan, Mudit Verma, K. Stechly, S. Bhambri, L. P. Saldyt, and A. B. Murthy. Position: Llms can’t plan, but can help planning in llm-modulo frameworks. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 22895–22907, 2024. URL <https://proceedings.mlr.press/v235/kambhampati24a.html>.
- [45] Mushfiqur R. Sarker, Matthew D. Murbach, Daniel T. Schwartz, and Miguel A. Ortega-Vazquez. Optimal operation of a battery energy storage system: Trade-off between grid economics and storage health. *Electric Power Systems Research*, 152:342–349, 2017. doi: 10.1016/j.epsr.2017.07.007.
- [46] Manajit Sengupta, Yu Xie, Anthony Lopez, Aron Habte, Galen Maclaurin, and James Shelby. The national solar radiation data base (nsrdb). *Renewable and Sustainable Energy Reviews*, 89:51–60, 2018. doi: 10.1016/j.rser.2018.03.003.
- [47] Kevin S. Anderson, Clifford W. Hansen, William F. Holmgren, Adam R. Jensen, Mark A. Mikofski, and Anton Driesse. pvlib python: 2023 project update. *Journal of Open Source Software*, 8(92): 5994, 2023. doi: 10.21105/joss.05994.
- [48] Eric Wilson, Andrew Parker, Anthony Fontanini, et al. End-use load profiles for the u.s. building stock, 2021.
- [49] Andrew Parker, Henry Horsey, Matthew Dahlhausen, Miles Praprost, Christopher CaraDonna, Andrew LeBar, and Lukas Klun. Comstock reference documentation (v.1). Technical Report NREL/TP-5500-83819, National Renewable Energy Laboratory, 2023.
- [50] California Independent System Operator. *Interface Specification for OASIS, Version 5.1.2*, 2017. URL https://www.caiso.com/documents/oasis-interfacespecification_v5_1_2clean_fall2017release.pdf.